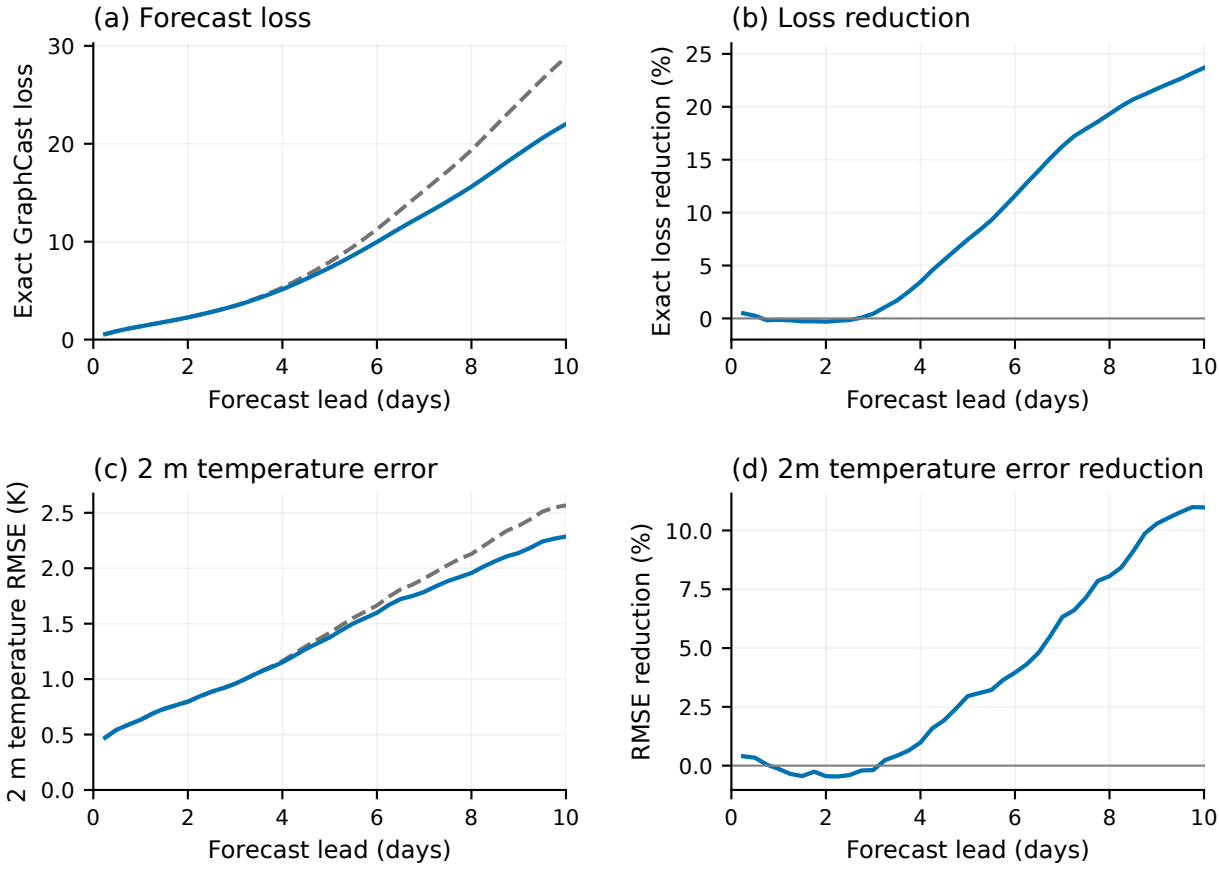
(a) Forecast loss
Exact GraphCast loss
Forecast lead (days)
(b) Loss reduction
Exact loss reduction (%)
Forecast lead (days)
(c) 2 m temperature error
2 m temperature RMSE (K)
Forecast lead (days)
(d) 2m temperature error reduction
RMSE reduction (%)
Forecast lead (days)
GraphCast baseline
3× Mamba LR + SWA
Mamba: $d_{\text{inner}} = 16$, one B, C group; SWA steps 500–2000